BEYOND HIGH PRICES: AUDITING INFORMATION-MEDIATED ENFORCEMENT IN RECURRENT PRICING AGENTS

SUPPLEMENTARY MATERIAL

Anonymous authors
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A EXTENDED INCENTIVE ANALYSIS

This supplement gives extended proofs, the executed PPO configuration, and all 160 conditional price-cut estimates. It reanalyzes the archived ten-seed sample in each condition; no additional policies were trained. The main manuscript frames the study as a causal audit of information-channel dependence, not as a replication of every Q-learning result in the algorithmic-collusion literature.

A.1 THE VALUE DIFFERENCE AND THE THRESHOLD

Consider the main text's stationary repeated game with absorbing, credible punishment. Write bounded cooperative, punishment, and specified one-period deviation payoffs as π^C, π^P, and π^D. Let $g = \pi^D - \pi^C$ and $\Delta V = \pi^C - \pi^P$. Assume $g > 0$ and $\Delta V > 0$. The signal depends on current actions; the deviator subsequently follows the prescribed automaton.

Subtract the punishment Bellman equation from the cooperative equation. Both values are evaluated before the current action. This timing gives

$$V^C - V^P = \frac{\pi^C - \pi^P}{1 - \delta}.$$

Since $\delta < 1$, the unique value difference is well defined. The deviation changes today's profit and the probability of entering punishment. Therefore

$$V^D - V^C = g - \delta(\beta - \alpha)(V^C - V^P).$$

Multiplying by the positive denominator preserves the inequality. The deviation is unprofitable if and only if

$$\delta \geq \frac{g}{g + (\beta - \alpha)\Delta V}.$$

Division yields δ^*. The strict feasibility condition $\delta^* < 1$ is equivalent to $\beta > \alpha$. Equality permits only $\delta = 1$, excluded here. If the threshold exceeds one, no admissible discount factor deters the specified deviation. A numerical threshold above one is therefore an infeasibility indicator, not an economically valid patience parameter.

A.2 WHY THE QUANTIFIER MATTERS

For each firm and action, compute its own current gain and detection probability. It is not enough to test the highest current-profit action. For example, take $\Delta V = 1$ and consider two deviations. An action with $g = 0.3$ and $\beta - \alpha = 0.5$ has threshold 0.588235. Another with $g = 0.9$ and $\beta - \alpha = 0.01$ has threshold 0.990099. At discount 0.90, the first is deterred and the second is not. These values are an algebraic illustration, not experimental estimates.

Main-text equation (10) must hold for every firm's deviation in C, with credible punishment after every history entering P. Neither this example nor the finite PPO probes establishes those requirements for all prices and histories.

A.3 Gaussian Noise and Its Boundaries

For a fixed threshold τ between μ_D and μ_C, let σ be the signal standard deviation. Define $z_C = (\tau - \mu_C)/\sigma$ and $z_D = (\tau - \mu_D)/\sigma$. The normal tail probabilities are $\alpha = \Phi(z_C)$ and $\beta = \Phi(z_D)$. Differentiation gives

$$\frac{\partial \Delta}{\partial \sigma} = -\frac{1}{\sigma}\big[z_D\phi(z_D) - z_C\phi(z_C)\big].$$

Combining these derivatives with the derivatives of δ^* gives

$$\frac{\partial \delta^*}{\partial \sigma} > 0.$$

Both channels of deterioration matter: more false punishment reduces the cooperative value, and lower detection reduces the expected loss caused by a deviation. As σ tends to zero, the threshold tends to 0. As σ tends to infinity, it tends to 2. If an additional independent Gaussian noise scale is σ_a, then $\sigma_{\text{tot}}^2 = \sigma^2 + \sigma_a^2$ and

$$\frac{\partial \delta^*}{\partial \sigma_a} = \frac{\partial \delta^*}{\partial \sigma_{\text{tot}}} \cdot \frac{\sigma_a}{\sigma_{\text{tot}}}.$$

For positive base noise, this derivative is positive at $\sigma_a > 0$ and zero at $\sigma_a = 0$. The threshold is still strictly higher at any positive added scale than at zero. This distinction prevents an incorrect claim of a strictly positive derivative at the boundary. The result holds the threshold, means, payoffs, and trigger strategy fixed. It does not sign the response of retrained PPO policies.

A.4 Observation Invariance and Finite Audit Horizons

Under the main text's action-independent channel, identical innovations and initial report queues produce identical reports even after an action intervention. Frozen actors receive only report histories and time. Their next prescribed actions therefore coincide; induction establishes equality at every later date. In a stationary payoff model, all later profit differences vanish. This proves the exact negative control without assuming any optimizer convergence.

The result would fail if actors received action-dependent sales, own-action memory, or rewards as inputs, or if the intervention changed a persistent economic state. During online learning, rewards could also change weights. The archived audit freezes weights, and the PPO execution inputs exclude those additional paths.

For a shorter audit, let H observed post-intervention periods remain within an episode ending at T, with intervention at t_0. If both branches' rewards lie in $[0, B]$, the absolute omitted discounted gain is bounded by

$$\left|\sum_{t=t_0+H+1}^{T} \delta^{t-t_0}(\pi_t^{\text{cut}} - \pi_t^{\text{ctrl}})\right| \le \frac{\delta^{H+1}B}{1-\delta}.$$

Here $B = 1.5$ is valid because price minus cost is at most 1.5 and demand is at most one. The archived probes use $H = T - t_0 - 1$, so this omitted finite-episode tail is zero. This does not bound deviations outside the probe family or establish immunity in an infinite-horizon game.

B Monitoring Calibration and Signal Timing

B.1 Parameterization of the Calibration Model

The native calibration model has six unconstrained coordinates θ. At the fixed threshold τ, it decodes them as

$$\mu_C = \tau + \exp(\theta_1), \quad \mu_D = \tau - \exp(\theta_2), \quad \sigma = \exp(\theta_3).$$

Independent normal priors on these coordinates have means $(0.5, 0.5, -0.5, 0, 0, 0)$ and standard deviations $(1, 1, 1, 2, 2, 2)$. The implementation rejects absolute values above 20 for the log coordinates and above 100 for the payoff location; thus the implemented target has numerical support bounds. The priors are defined on the coordinates actually sampled. No change-of-variables Jacobian is added to that coordinate-space target.

Table 1: Signal generation and delivery sequence within one market period.

Stage	Operation
1	Both firms act using the existing 32-report window and time.
2	The C++ core maps actions to prices and computes demand and profits.
3	The minimum price determines the clipped latent monitoring mean.
4	The last two available latent means are averaged; one normal innovation is added.
5	The generated report enters a one-element queue; the oldest report is delivered.
6	The delivered report is normalized and appended to the next observation.

Labeled cooperative and deviating signals have normal likelihoods with means μ_C and μ_D and common scale σ. Labeled benchmark payoffs also use normal likelihoods, with a supplied known payoff observation scale. These labels and restrictions are model inputs. The posterior does not discover a cartel label from an unlabeled price series.

Only (μ_C, μ_D, σ) are consumed by the PPO posterior sampler. Their joint rows are sampled with replacement, preserving the empirical cross-coordinate dependence, and held fixed throughout an episode. The payoff coordinates and derived theoretical critical discounts are not PPO rewards. Profits are recomputed from logit demand and realized prices. The archived manifest identifies the posterior by hash and records a passed diagnostic gate; it does not contain enough raw calibration evidence to establish external validity.

B.2 Timing in the Executed Channel

A price change at time t first changes the generated report at t. With delay one, that report is delivered after the action at $t + 1$, so the earliest actor response is at $t + 2$. This accounts for the initial zero entries of the informative impulse response. Overlapping two-period latent-mean windows can spread the signal effect across reports.

The ablation replaces the generated report by $r_{t+1} \sim \mathcal{N}(1.6, 0.4^2)$. It retains shared randomness: both firms see the same signal, so the experiment removes information about actions, not all common randomization. Identical innovation consumption across paired branches maintains common-random-number alignment. The retained common signal can correlate actions even when it cannot reveal the forced deviation.

C PPO Implementation and Executed Configuration

The two firms use distinct actor–critic parameters and Adam optimizers. The GRU choice follows the observation structure rather than a claim about collusion: the public report is delayed, aggregated, and noisy, so one observation cannot determine whether a price movement is an isolated fluctuation or a deviation. A gated recurrent unit (Cho et al., 2014) provides a compact finite-memory map over the report history. Its update and reset gates can retain, attenuate, or combine evidence across time. Each encoder runs this GRU from a fresh hidden state across the 32-observation window, concatenates normalized time, then applies a linear projection, $\tanh$, and layer normalization. Thus the implementation is a windowed recurrent encoder, not an unrestricted episode-level memory state. Policy and value heads share that encoder within a firm. There is no shared actor between firms and no centralized critic in the PPO path. The retaliatory mechanism studied here requires separate evidence of elevated prices, action-contingent punishment, and deterrence. Other coordination mechanisms need not involve punishment. No comparison with an MLP on the same window or a CNN establishes GRU superiority.

Advantages are standardized within the complete rollout separately for each firm, using the population standard deviation plus ϵ. For likelihood ratio r_t and standardized advantage $\hat{A}_t$, the minimized loss is

$$L = -\min(r_t \hat{A}_t, \mathrm{clip}(r_t, 1-\varepsilon, 1+\varepsilon)\hat{A}_t) + c_v \cdot \tfrac{1}{2}\mathrm{MSE} - c_e H.$$

The effective MSE coefficient is 0.25 because the code multiplies MSE by 0.5 and then applies value coefficient 0.5. Entropy is estimated for the bounded $\tanh$ policy, not its base normal. Stored pre-$\tanh$ actions avoid inverse-$\tanh$ instability.

Table 2: Configuration recorded in the completed job files.

Parameter	Executed Value
Training seeds in each condition	20261001 through 20261010
Training interactions per run	1 000 000 joint market transitions
Parallel environments and episode length	2 environments; 512 periods
Public history and encoder width	32 reports; 128 GRU units
Full rollout size	2048 periods per environment; 4096 transitions
Optimization epochs and minibatch	10; 256
Adam learning rate and epsilon	0.0003; 0.000 01
Discount and GAE parameter	0.99; 0.95
PPO clipping and entropy coefficient	0.2; 0.01
Value coefficient and gradient norm cap	0.5; 0.5
Initial actor log std. dev.	-0.5; clamped to $[-5, 2]$ during use
Informative channel	Delay 1; aggregation 2; added noise scale 0.1
Action-independent channel	Same delay; report $\sim \mathcal{N}(1.6, 0.4^2)$
Diagnostics and final evaluation	100 checkpoints $\times$ 16 episodes; final 64 episodes
Probe family and paired repetitions	2 firms $\times$ 2 times $\times$ 2 cuts; 128 pairs per probe

Table 3: Final evaluation of every completed PPO policy pair. Each row averages 64 evaluation episodes. B denotes informative baseline and A denotes action-independent retraining. Seed suffix 01 denotes 20261001, continuing through 10. Rank columns describe projected, time-augmented encoder features.

Seed	Arm	Price	Elevation	Surplus	Rank 0	Rank 1
01	B	1.497 102	0.053 323	1.157 867	1.276 308	1.237 626
01	A	1.488 946	0.035 278	1.159 020	1.418 177	1.386 605
02	B	1.481 101	0.017 922	1.159 558	1.218 294	1.241 206
02	A	1.476 651	0.008 077	1.160 178	1.547 627	1.522 354
03	B	1.468 815	−0.009 260	1.161 126	1.229 325	1.311 140
03	A	1.458 835	−0.031 339	1.162 020	1.328 914	1.342 101
04	B	1.479 467	0.014 307	1.160 148	1.254 956	1.291 669
04	A	1.477 020	0.008 893	1.160 382	1.232 180	1.275 160
05	B	1.509 558	0.080 881	1.155 365	1.281 766	1.237 738
05	A	1.466 939	−0.013 408	1.161 483	1.331 038	1.422 554
06	B	1.461 530	−0.025 377	1.160 801	1.259 499	1.229 360
06	A	1.487 097	0.031 188	1.159 354	1.293 448	1.355 741
07	B	1.483 384	0.022 973	1.159 627	1.265 380	1.228 610
07	A	1.493 508	0.045 372	1.158 365	1.280 998	1.314 625
08	B	1.463 500	−0.021 018	1.161 653	1.342 083	1.293 155
08	A	1.493 939	0.046 324	1.157 274	1.310 864	1.304 309
09	B	1.474 509	0.003 338	1.159 803	1.280 268	1.268 246
09	A	1.491 027	0.039 883	1.158 652	1.291 892	1.254 833
10	B	1.479 070	0.013 430	1.159 930	1.335 473	1.356 627
10	A	1.480 958	0.017 607	1.159 871	1.351 331	1.357 494

Episode termination has zero bootstrap; nonterminal rollout boundaries use the next-state value. A shortened final rollout reaches exactly one million interactions. Environments reset synchronously. Deployment uses $\tanh$ of the latent mean; stochastic-policy performance is not reported.

D Complete Final Outcomes by Training Seed

Between-seed price standard deviations are 0.014713 for B and 0.011766 for A. Elevation standard deviations are 0.032551 and 0.026032. A mean near the Nash reference does not rule out asymmetry, variation across time, or isolated profitable price cuts. Negative elevation is retained as a valid measured outcome rather than truncated to zero.

The reported effective-rank columns are descriptive diagnostics of the projected, time-augmented encoder features. They do not measure monitoring quality, whether the firms learned the same representation, or whether a punishment rule was encoded. The GRU is a finite-memory window encoder, not evidence of a persistent collusive memory state. A rank difference therefore cannot be interpreted as a causal channel effect without additional representation and policy interventions.

Table 4: Seed-bootstrap intervals for cut-specific full gain and its continuation component. Intervals are pointwise and unadjusted.

Arm	Cut	Quantity	Mean	Lower 95%	Upper 95%
B	0.1	Full gain	−0.005 192	−0.005 855	−0.004 542
B	0.1	Continuation	−0.000 005	−0.000 010	−0.000 001
B	0.3	Full gain	−0.051 322	−0.053 372	−0.049 286
B	0.3	Continuation	−0.000 005	−0.000 010	−0.000 001
A	0.1	Full gain	−0.004 800	−0.005 393	−0.004 343
A	0.1	Continuation	0.000 000	0.000 000	0.000 000
A	0.3	Full gain	−0.050 273	−0.052 023	−0.048 937
A	0.3	Continuation	0.000 000	0.000 000	0.000 000

All 20 rows are included in the reported comparison. A final 64-episode estimate differs slightly from the last 16-episode learning-curve estimate because their evaluation streams differ. The two quantities should not be silently substituted for one another.

E Statistical Estimands and Uncertainty

Let Y_s^B and Y_s^A denote final outcomes for paired training seed s. We estimate the mean channel-replacement effect as

$$\hat{\Delta} = \frac{1}{10} \sum_{s=1}^{10} (Y_s^A - Y_s^B).$$

Each bootstrap draw resamples ten paired seed indices with replacement. We report the 2.5th and 97.5th percentiles from 2000 draws using NumPy `default_rng`, seed 20260920. Whole-seed resampling also gives pointwise learning-curve bands. Intervals condition on the posterior, simulator, hyperparameters, and evaluation protocol; they are not simultaneous bands.

For audits, average probe types within each seed before resampling seeds. Cut-specific means include four types; overall rival-response area includes eight. Seeds receive equal weight; probe repetitions are not independent training runs.

E.1 Conditional Intervals in the Complete Probe Tables

Each individual probe file also records a 1000-resample bootstrap interval for the mean across its 128 paired rollouts. The bootstrap generator is initialized with the audit seed plus 54 321. Tables 5–**??** report those stored mean intervals. Their interpretation is conditional on one trained policy pair. The separately stored rollout quantiles named `deviation_gain_q025` and `deviation_gain_q975` are not confidence limits for a mean and are not used in those tables.

In Tables 5–**??**, seed suffixes 01–10 identify 20261001–20261010; t is intervention time. Immediate denotes impact gain; G includes the remaining episode. Area sums the rival's mean price response after impact, in price-period units. Zeros are rounded except exact action-independent responses. Lower and upper limits are unadjusted conditional mean intervals.

Within-seed probes share policies and random-stream construction. Fixed diagnostic streams across checkpoints likewise do not create independent observations. The 100 checkpoints cannot be treated as 100 independent training samples.

E.2 What an Unresolved Difference Means

The price-elevation interval includes zero and effects of either sign. Without a tested equivalence margin, this does not establish equivalence. The four positive conditional small-cut means are descriptive, not multiplicity-adjusted discoveries. Reporting both averages and individual probes preserves this heterogeneity.

F Complete Probe Tables

Table 6: Archived cryptographic hashes for reproducibility verification.

Component	SHA256
Posterior draws	`a2cd4f268b3edf0d...202d1b95`
Native extension	`8c1fe82a4830d09e...ef882780`
PPO training source	`12ffcf180b6fc899...662d3f15`

Table 5: Normalized cut 0.1 under informative baseline. Unclipped price cut 0.075. Each row uses 128 paired rollouts.

Seed Area	Firm	t	Immediate	G	Lower 95%	Upper 95%
01 -9.91e-05	0	64	0.000 613	0.000 573	0.000 557	0.000 591
01 -4.23e-05	0	256	−0.001 242	−0.001 259	−0.001 284	−0.001 234
01 -3.63e-05	1	64	−0.007 589	−0.007 606	−0.007 633	−0.007 581
01 -1.54e-05	1	256	−0.007 374	−0.007 381	−0.007 422	−0.007 339
02 1.28e-07	0	64	−0.008 006	−0.008 006	−0.008 023	−0.007 989
02 0.00e+00	0	256	−0.009 356	−0.009 356	−0.009 381	−0.009 328
02 -1.36e-07	1	64	−0.000 680	−0.000 680	−0.000 701	−0.000 658
02 0.00e+00	1	256	−0.001 313	−0.001 313	−0.001 346	−0.001 279
10 0.00e+00	1	256	−0.002 912	−0.002 912	−0.003 033	−0.002 792

Column definitions and conditional-interval limitations appear in Section E.1.

G Reproducibility and Computation Record

G.1 Provenance of the Reported Outputs

The results archive contains complete final PPO checkpoints, job configurations, learning curves, probe summaries, and suite metadata. The archived Python source hashes and all 20 final checkpoint hashes were checked against the manifest and job files when preparing the main text. The executable binary itself is not distributed in this results archive; its recorded hash alone does not establish that a newly compiled binary is byte-identical.

The recorded environment uses Python 3.13.15, NumPy 2.1.3, PyTorch 2.11.0+cu128, Gymnasium 1.3.0, and device `cuda`. These are archived versions, not a new compatibility guarantee.

The diagnostic gate records maximum rank-normalized split $\hat{R} = 1.008945$ and minimum bulk/tail ESS $= 606.86$. Passing thresholds of 1.01 and 400 addresses sampling diagnostics, not calibration-model adequacy. This supplement does not reconstruct missing likelihood inputs or invent posterior moments from those two statistics. Independent calibration replication requires the original labeled observations, observation-scale settings, sampling configuration, and retained draws matching the recorded posterior hash.

G.2 Interaction Accounting

Each audit type costs $131\,072$ transitions, including the pre-intervention prefix in both branches. Eight types give the reported $1\,048\,576$. One market transition contains both firms' actions; it is not two separate environment interactions. Recorded completed-job times sum to 21.978 hours for baseline and 22.079 hours for ablation.

Three previous-attempt records concern baseline seed 20261010 and ablated seeds 20261008 and 20261009. All are marked running, not failed, in the archived records. Their filenames contain `previous_failure`, but

Table 7: Recorded interaction counts across all completed jobs.

Component	Per Job	Across 20 Jobs
Training	1 000 000	20 000 000
Training diagnostics	819 200	16 384 000
Trainer final eval.	8192	163 840
Suite final eval.	32 768	655 360
Paired deviation audit	1 048 576	20 971 520
Total recorded	2 908 736	58 174 720

Table 8: Unresolved extensions and the questions they would address.

Additional Experiment	Question It Would Resolve
Known collusive trigger as positive control	Does the audit detect a designed gain/loss tradeoff?
Calvano-style tabular Q-learning benchmark	Does the implementation reproduce prior results?
Matched-moment action-independent reports	Is the effect due to information or input scale?
Nonsaturating signal maps and smaller cuts	Does saturation near Nash conceal sensitivity?
Recurrent versus memoryless policies	Does history processing change enforcement?
Adaptive and multistep deviations	Do policies admit profits outside fixed probes?
Continuing-task and horizon sensitivity	How much do terminal date and finite memory matter?
Gaussian-noise sweep with fixed trigger	Does the comparative static hold empirically?
Alternative learners and market primitives	How far do PPO findings generalize?

the records do not establish a numerical training failure or its cause. All three final jobs completed. Unrecorded interrupted work is excluded from the interaction and runtime totals above.

H Reproduction Steps and Unresolved Extensions

H.1 Reproduce the Reported PPO Scope

Use the archived source and build instructions. Supply the original `posterior_draws.csv` and matching `posterior_summary.csv`, verify the posterior hash, and run `require_converged`. Preserve joint posterior rows; independent marginal sampling changes the training distribution.

Copy `experiments/research.json`, retaining `ppo_baseline`, `ppo_ablated`, and their paired comparison. Keep seeds, evaluation counts, and audit settings unchanged. Save as `experiments/ppo_research.json`. The original suite also scheduled SAC; this PPO-only recipe changes plan identity and requires a new output directory.

Run `python -m acrl.experiment_suite` with `--plan experiments/ppo_research.json`, `--posterior results/mcmc/posterior_draws.csv`, `--output results/ppo_reproduction`, and `--device cuda`. CPU changes run identity and may change floating-point behavior. Exact replication across hardware or package versions requires verification.

The `--resume` option reuses validated completed jobs with matching identities. It does not continue an interrupted optimizer state. A progress policy checkpoint alone does not restore all optimizer and random-generator states. Preserve the prior archive and identify restarted jobs.

Check job statuses, hashes, and finite metrics. Recompute seed summaries from `job.json`, curves from `policy.curve.csv`, and probes from `audit_probes.json` using Section E's conventions. Exclude unfinished SAC runs.

H.2 Extensions That Would Strengthen the Evidence

The following are not completed experiments. They define limits of the conclusions. A Calvano-style Q-learning replication provides an external benchmark and can serve as an empirical positive control if its enforcement is independently verified. A designed trigger strategy supplies a control with a known current-gain/continuation-loss tradeoff.

REFERENCES

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